Problem Set 16

Problem 13.3.1: To say the pion has a range of $\lambda \simeq 10^{-5}$ angstroms is to say that a single pion can be localized on this scale: $\Delta X \sim \lambda$. From the uncertainty principle $\Delta X\,\Delta P \gtrsim \hbar$, we then deduce $\Delta P \gtrsim \hbar/\lambda$. Since we assume λ is the smallest scale on which we can localize the pion, it is plausible that the inequality is saturated, so $P \sim \Delta P \sim \hbar/\lambda$. The relation between the energy and momentum is (from special relativity) $E^2 = m^2c^4 + c^2P^2 \sim m^2c^4 + c^2\hbar^2/\lambda^2$. Now, for the notion of a "single pion" to exist, we must have $E \lesssim 2mc^2$. (See discussion in text on p. 363). So $4m^2c^4 \gtrsim m^2c^4 + c^2\hbar^2/\lambda^2$, or $mc^2 \gtrsim c\hbar/(\sqrt{3}\lambda)$. Again, since λ is the smallest scale, it is plausible that the inequality is saturated, giving

$$mc^2 \sim \frac{c\hbar}{\sqrt{3}\lambda} \sim \frac{2000\,\mathrm{eV}\ \mathring{A}}{1.7\times 10^{-5}\ \mathring{A}} \sim 100\,\mathrm{MeV}.$$

(In reality $m_\pi c^2 = 140$ MeV.)

Problem 13.3.2: Since the kinetic energy is $T = 200$ eV $\ll 0.5$ MeV $\simeq mc^2$, the electron is non-relativistic, so we can use $T = p^2/(2m) = (p^2c^2)/(2mc^2)$ which implies $pc = \sqrt{2mc^2T}$. Then

$$\lambda = \frac{2\pi\hbar}{p} = \frac{2\pi\hbar c}{pc} = \frac{2\pi\hbar c}{\sqrt{2mc^2T}} \simeq \frac{\sqrt{2}\pi(2000\,\mathrm{eV}\ \mathring{A})}{\sqrt{(0.5\times 10^6\mathrm{eV})(200\,\mathrm{eV})}} = \frac{\sqrt{2}\pi}{10}\ \mathring{A} \simeq 1\ \mathring{A}\,.$$

Problem 13.3.3: Recalling $E_n = -\mathrm{Ry}/n^2 \simeq -13\mathrm{eV}/n^2$, we have

$$\frac{P(n=2)}{P(n=1)} = 4e^{-(E_2-E_1)/kT} = 4e^{-[(-1/4)-(-1)]13\,\mathrm{eV}/(k_BT)} \simeq 4e^{-10^5/(T/K)}$$

where I used $(k_B/\mathrm{eV}) \sim 9\times 10^{-5}\mathrm{K}^{-1}$. So it is clear that we need $T \gtrsim 10^5$ K so that the exponent is not very small. For example, if $T = 6000$ K, then

$$\frac{P(n=2)}{P(n=1)} \simeq 4e^{-10^5/(6000)} \simeq 4e^{-16} \sim 2\times 10^{-7} \ll 1.$$